



BobSim

SteadyStateEval Vehicle Characterization

Measured- a_y ramp-steer isolines with robust fits

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Run Notes

- Open-loop ramp-steer at 12.5, 15, 17.5, and 20 m/s
- steerStart = 2.0 s; trim all pre-ramp samples
- Measured a_y is the analysis axis; commanded a_y only sets the ramp
- lmeasured a_y < 1.0 m/s² is excluded; linear band = 1.0 to 4.0 m/s²
- Understeer gradient = mean $d(\text{delta_excess})/d(a_y)$ over the linear band
- Response curves use split Hampel-filtered spline fits; sensitivities trim the outer 20% of lmeasured a_y
- steerExcess = avgSteerAngle - wheelbase * curvature
- Tire/config: MF52, baseline_v3; Backend: OpenModelica

SteadyStateEval Summary 12.5 m/s

Metric	Value	Units
Measured a_y Range	1.07 → 17.40	m/s^2
Roadwheel Angle Gradient	0.01034	$\frac{\text{rad}}{\text{m/s}^2}$
	5.81	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.04245	$\frac{\text{rad}}{\text{m/s}^2}$
	23.86	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00210	$\frac{\text{rad}}{\text{m/s}^2}$
	1.18	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00048	$\frac{\text{rad}}{\text{m/s}^2}$
	0.27	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.04245	$\frac{\text{rad}}{\text{m/s}^2}$
	23.86	$\frac{\text{deg}}{g}$
Roll Gradient	1.443	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-18.2	$\text{N} \cdot \text{m}$
Handwheel Torque Max	-1.9	$\text{N} \cdot \text{m}$

SteadyStateEval Summary 15.0 m/s

Metric	Value	Units
Measured a_y Range	1.07 $\rightarrow$ 18.07	m/s ²
Roadwheel Angle Gradient	0.00754	$\frac{\text{rad}}{\text{m/s}^2}$
	4.24	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.03096	$\frac{\text{rad}}{\text{m/s}^2}$
	17.40	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00083	$\frac{\text{rad}}{\text{m/s}^2}$
	0.47	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00068	$\frac{\text{rad}}{\text{m/s}^2}$
	0.38	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.03096	$\frac{\text{rad}}{\text{m/s}^2}$
	17.40	$\frac{\text{deg}}{g}$
Roll Gradient	1.441	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-17.5	N · m
Handwheel Torque Max	-1.9	N · m

SteadyStateEval Summary 17.5 m/s

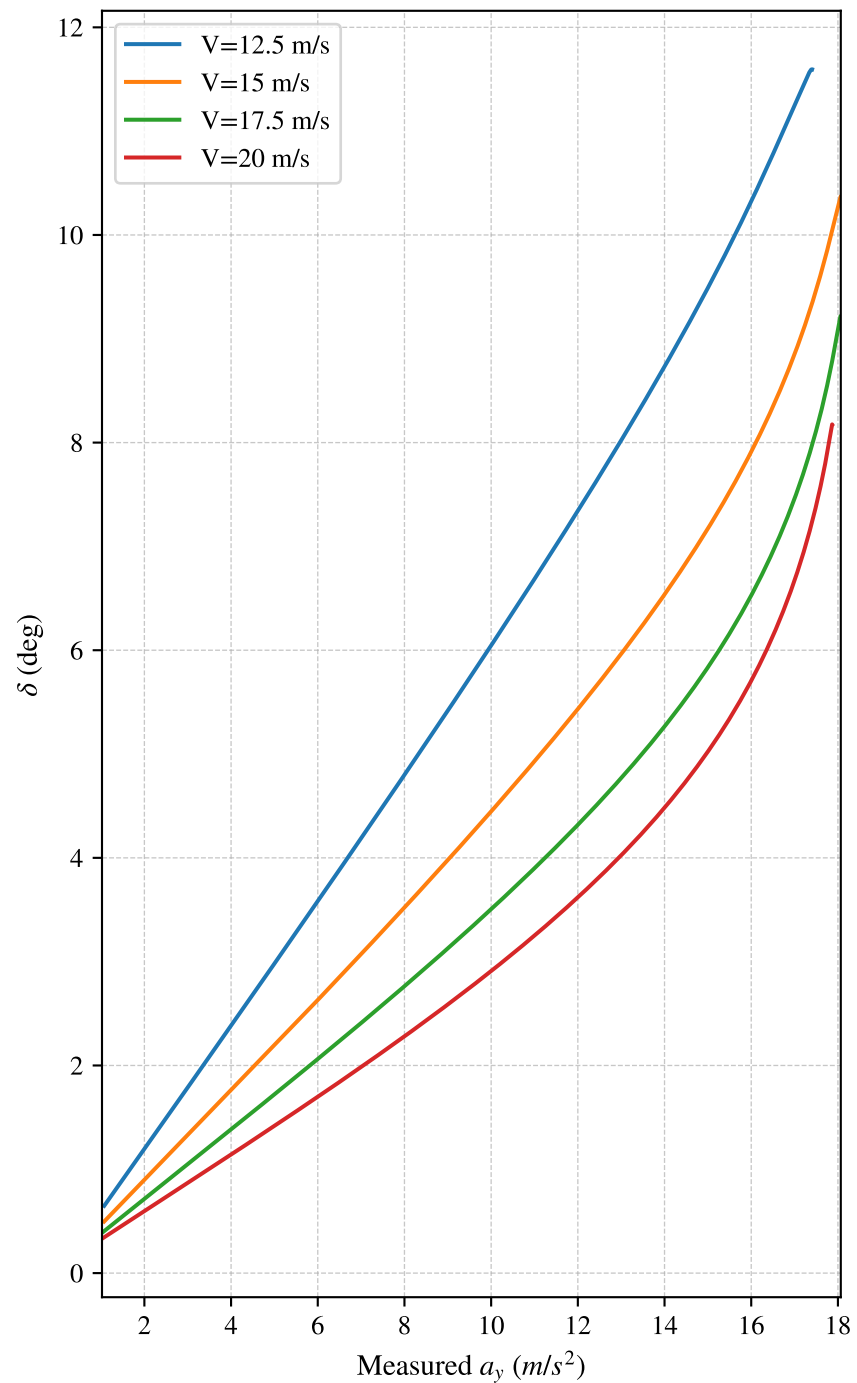
Metric	Value	Units
Measured a_y Range	1.05 $\rightarrow$ 18.06	m/s ²
Roadwheel Angle Gradient	0.00587	$\frac{\text{rad}}{\text{m/s}^2}$
	3.30	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.02411	$\frac{\text{rad}}{\text{m/s}^2}$
	13.55	$\frac{\text{deg}}{g}$
Sideslip Gradient	-0.00004	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.02	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00072	$\frac{\text{rad}}{\text{m/s}^2}$
	0.40	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.02411	$\frac{\text{rad}}{\text{m/s}^2}$
	13.55	$\frac{\text{deg}}{g}$
Roll Gradient	1.445	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-17.0	N · m
Handwheel Torque Max	-1.9	N · m

SteadyStateEval Summary 20.0 m/s

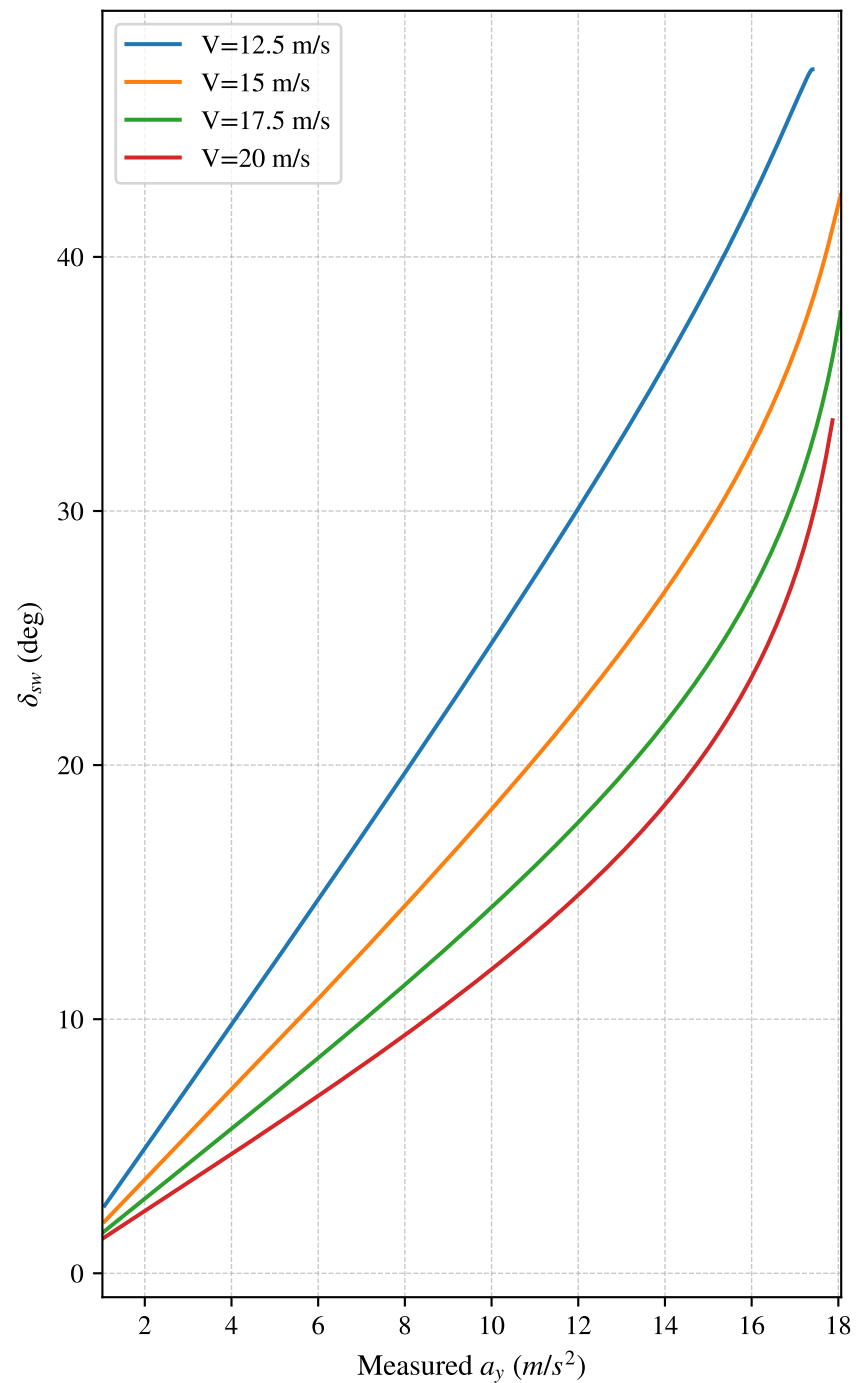
Metric	Value	Units
Measured a_y Range	1.02 → 17.87	m/s^2
Roadwheel Angle Gradient	0.00477	$\frac{\text{rad}}{\text{m/s}^2}$
	2.68	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.01962	$\frac{\text{rad}}{\text{m/s}^2}$
	11.03	$\frac{\text{deg}}{g}$
Sideslip Gradient	-0.00041	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.23	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00079	$\frac{\text{rad}}{\text{m/s}^2}$
	0.45	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.01962	$\frac{\text{rad}}{\text{m/s}^2}$
	11.03	$\frac{\text{deg}}{g}$
Roll Gradient	1.454	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-16.6	$\text{N} \cdot \text{m}$
Handwheel Torque Max	-1.9	$\text{N} \cdot \text{m}$

Steering Angle Isolines vs Measured a_y

Roadwheel Angle δ

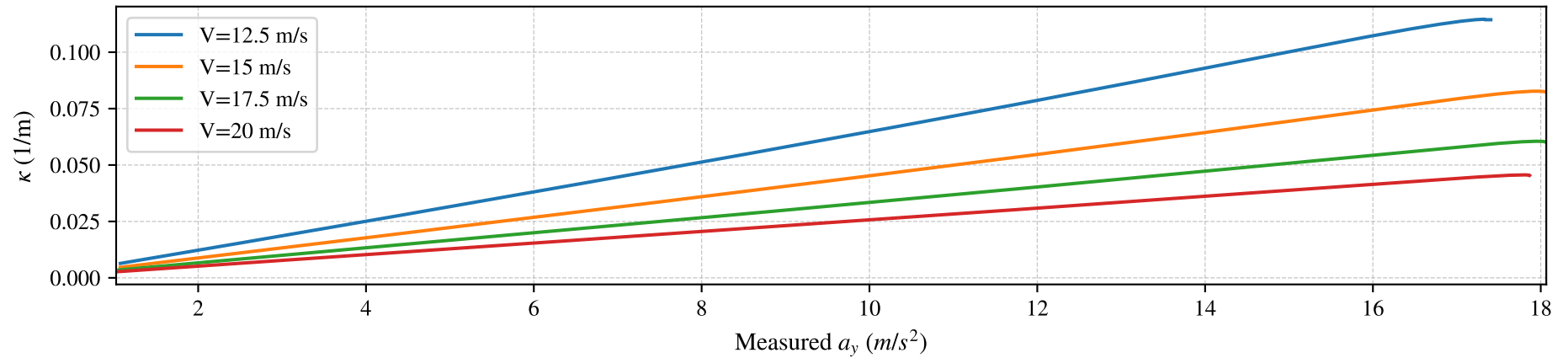


Handwheel Angle δ_{sw}

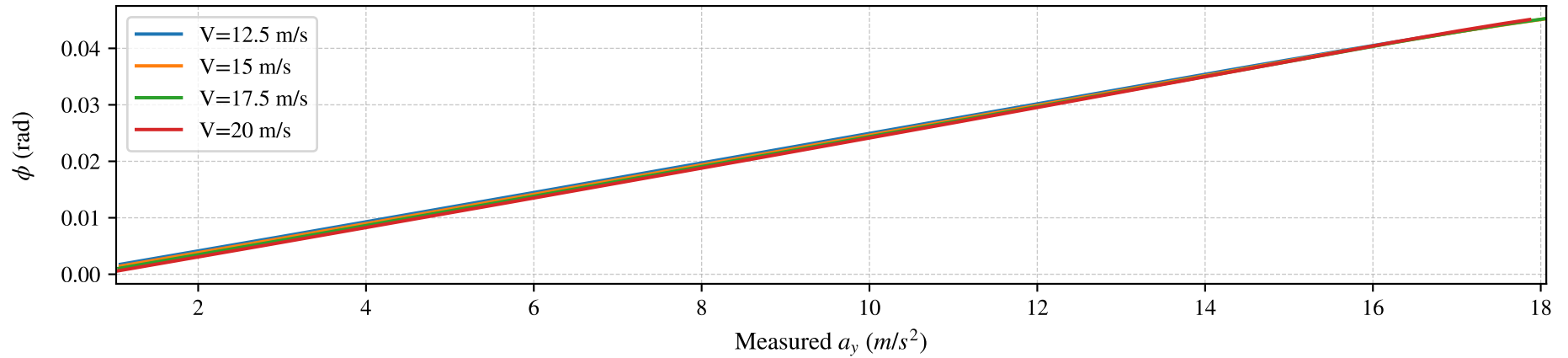


Vehicle State Isolines vs Measured a_y

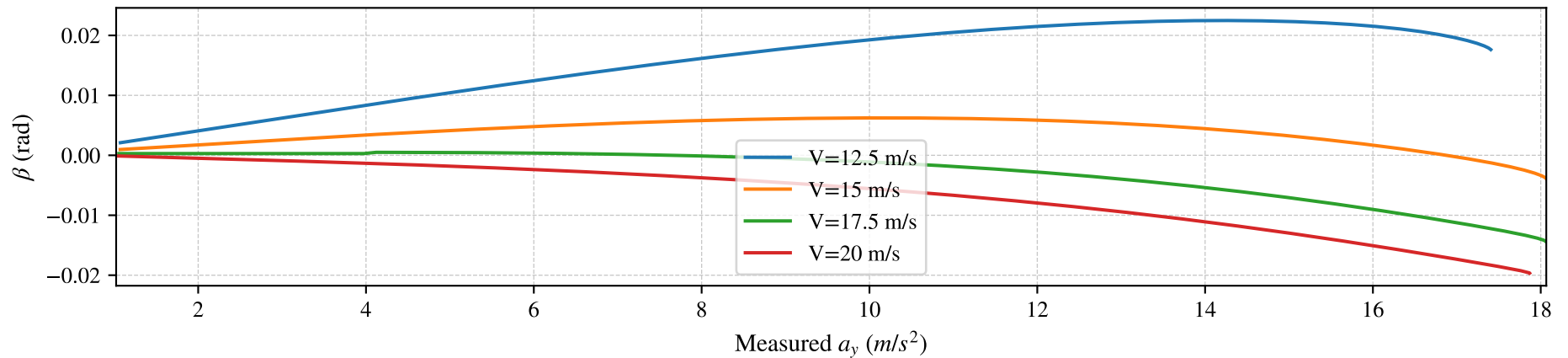
Curvature κ



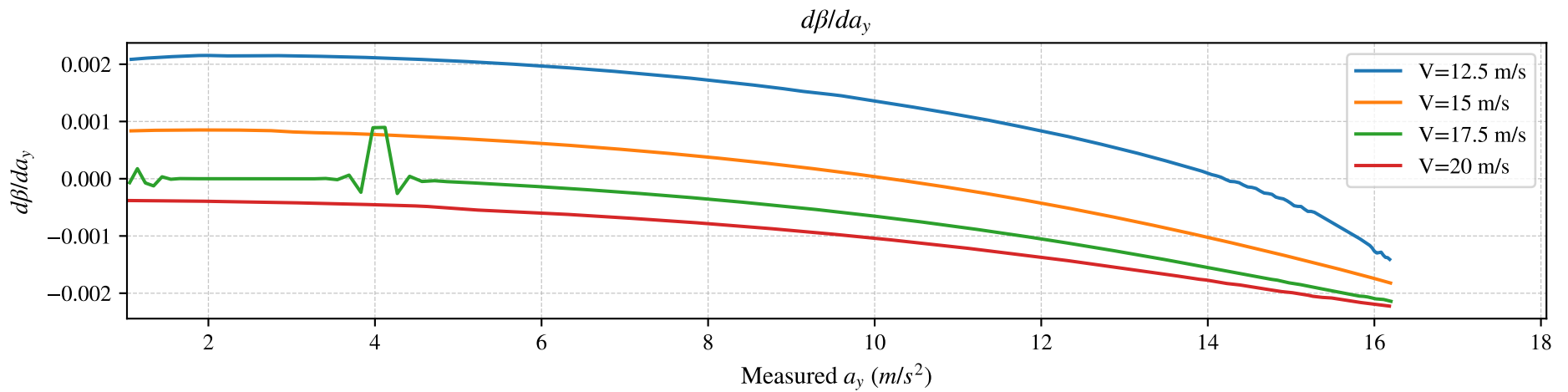
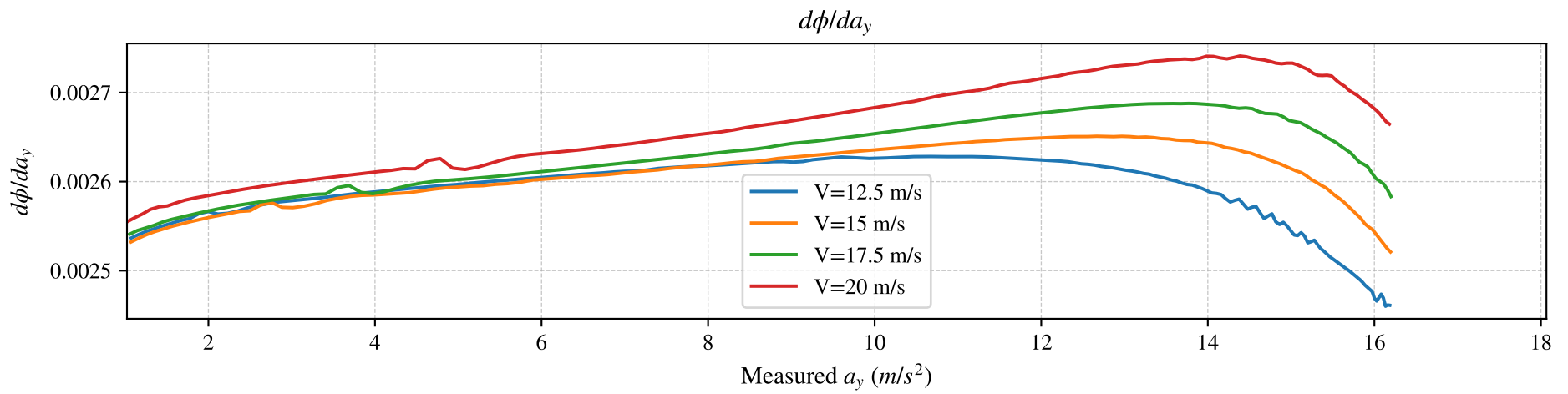
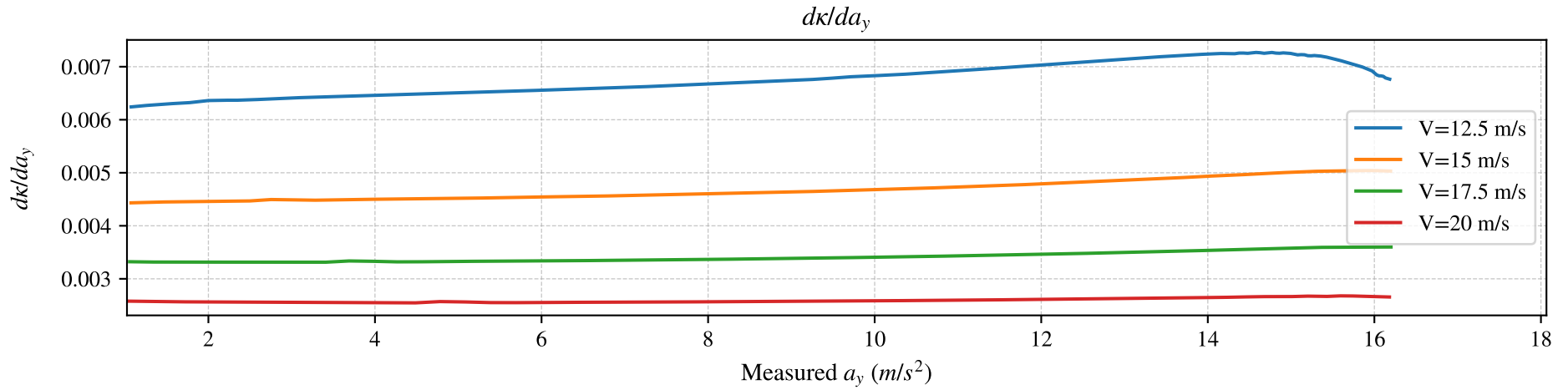
Roll Angle ϕ



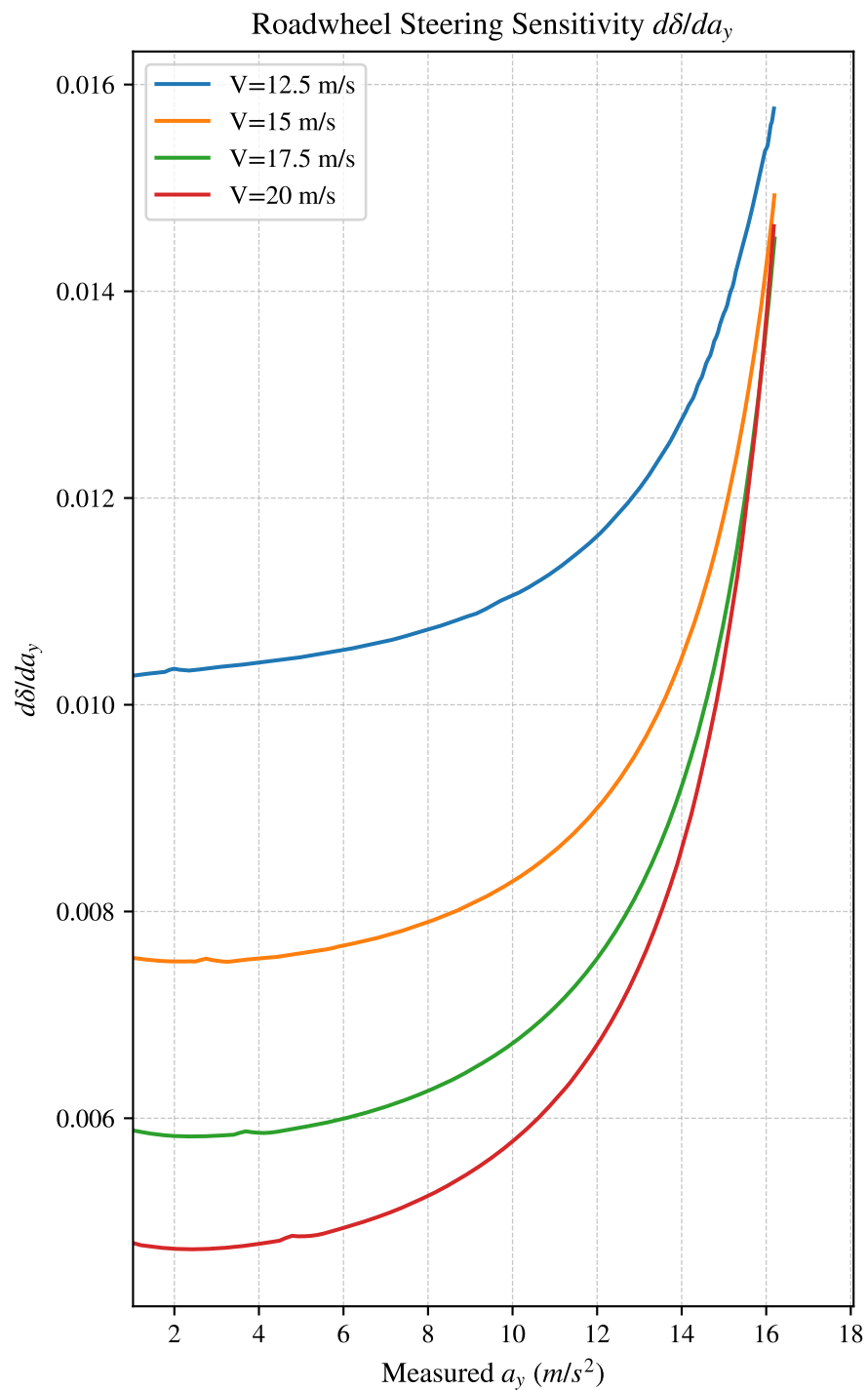
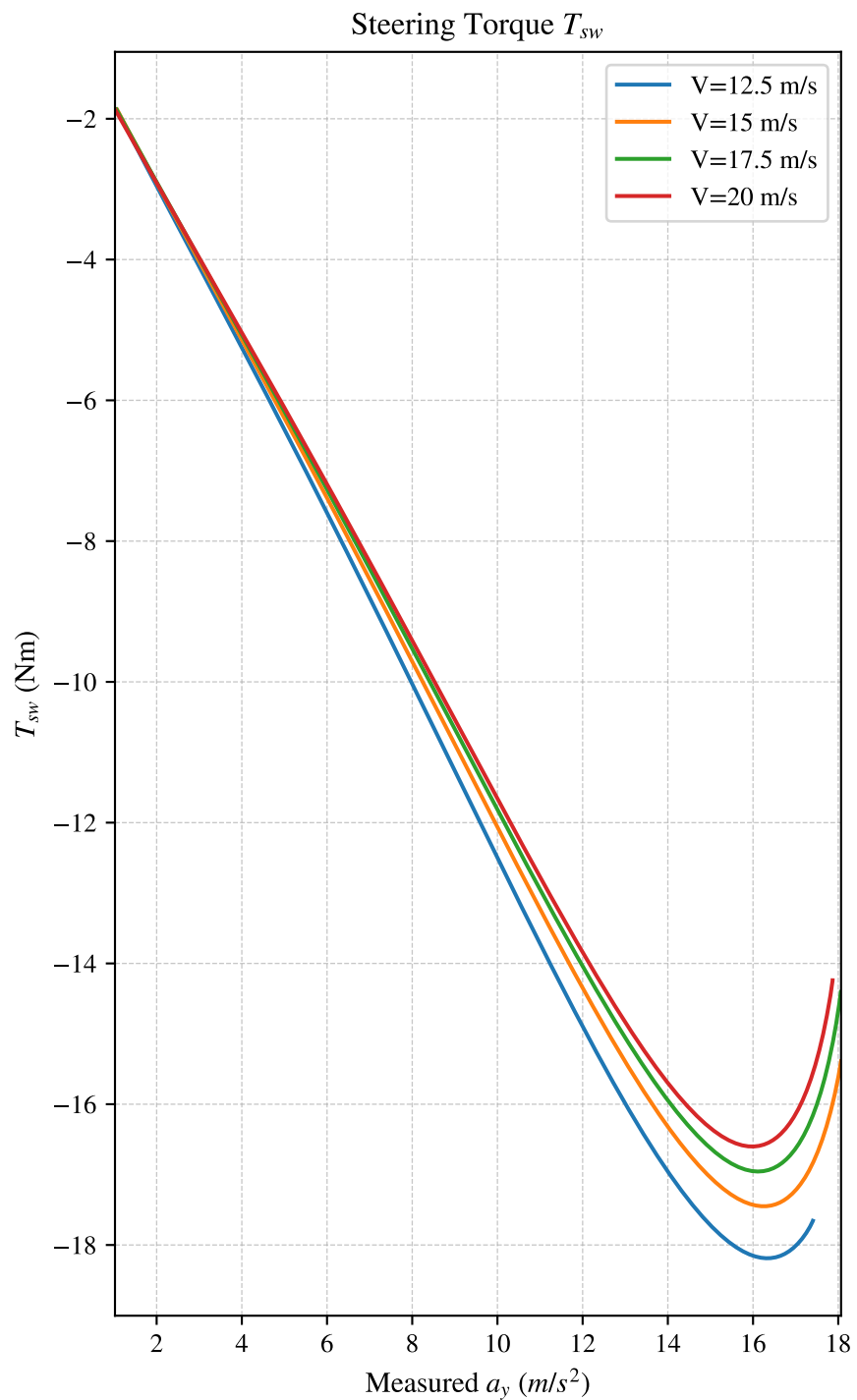
Sideslip β



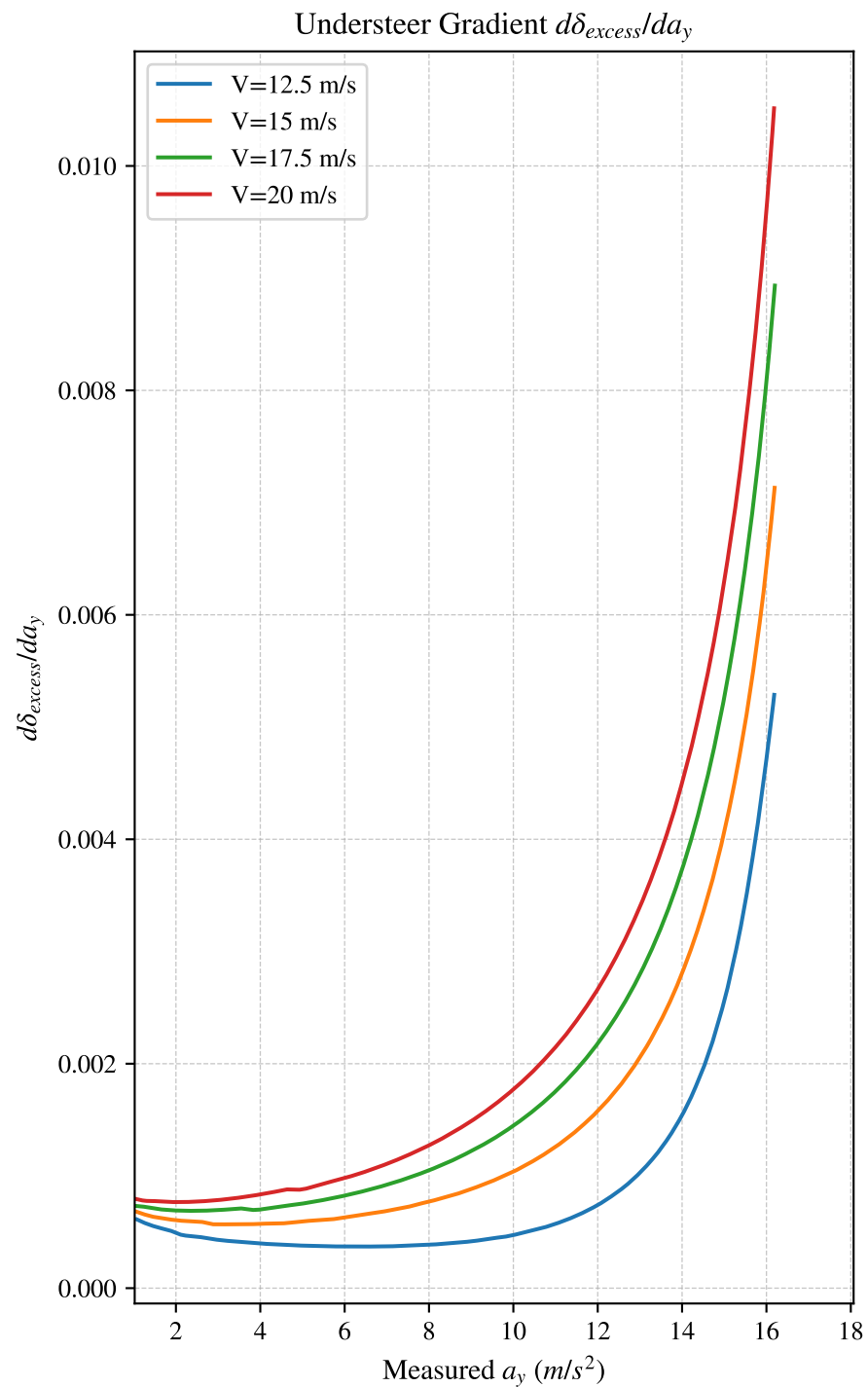
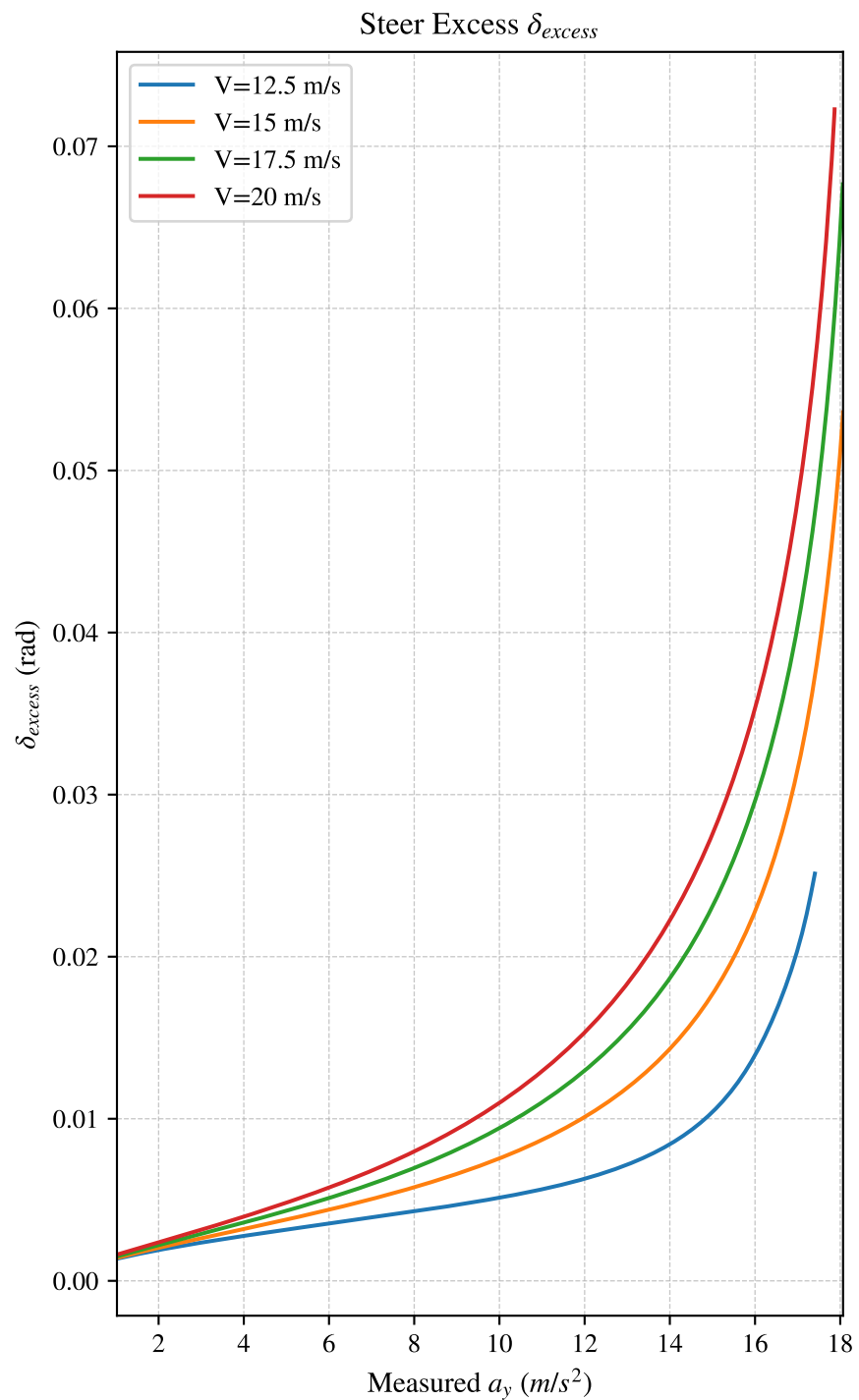
Parametric Response Sensitivities vs Measured a_y



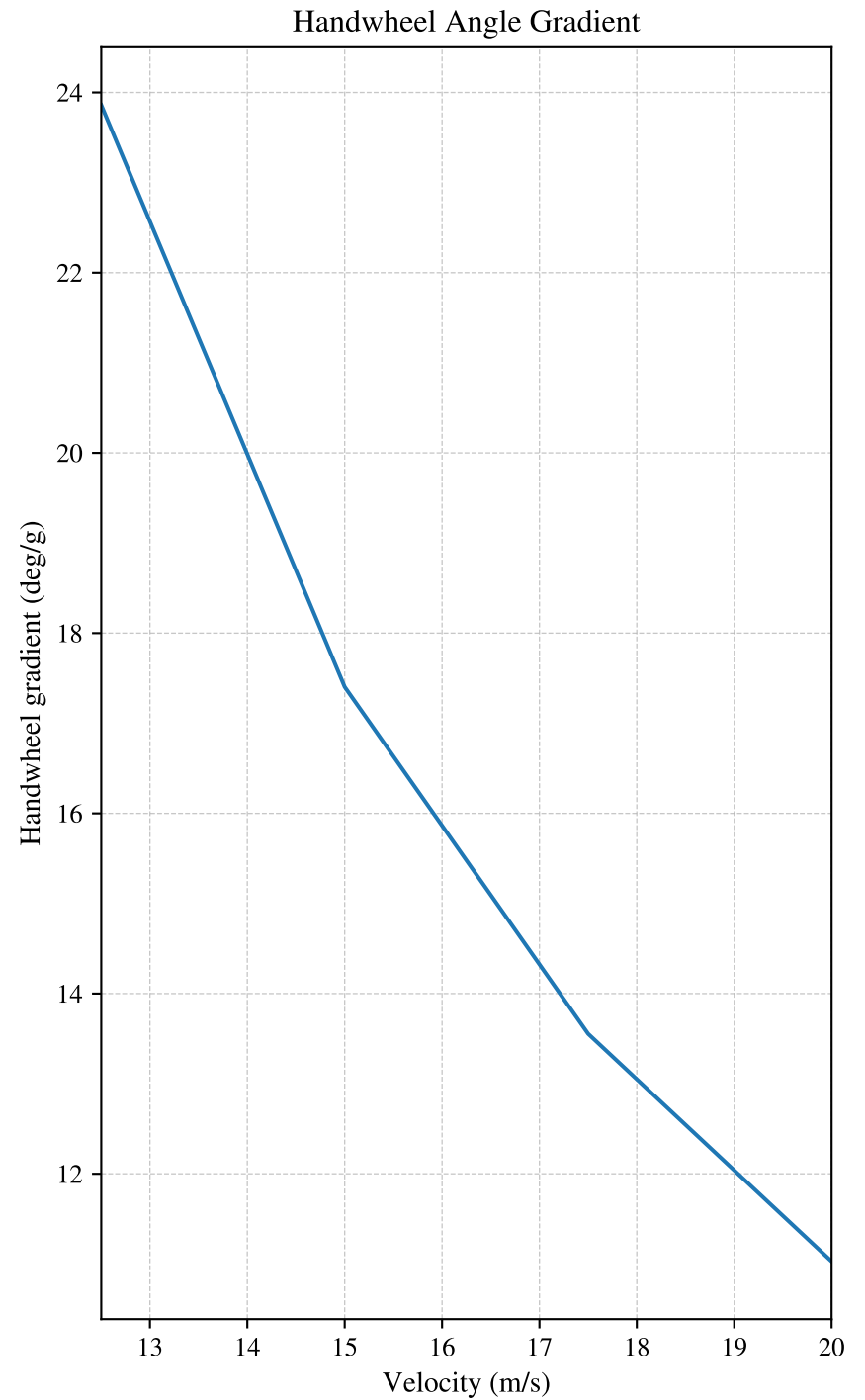
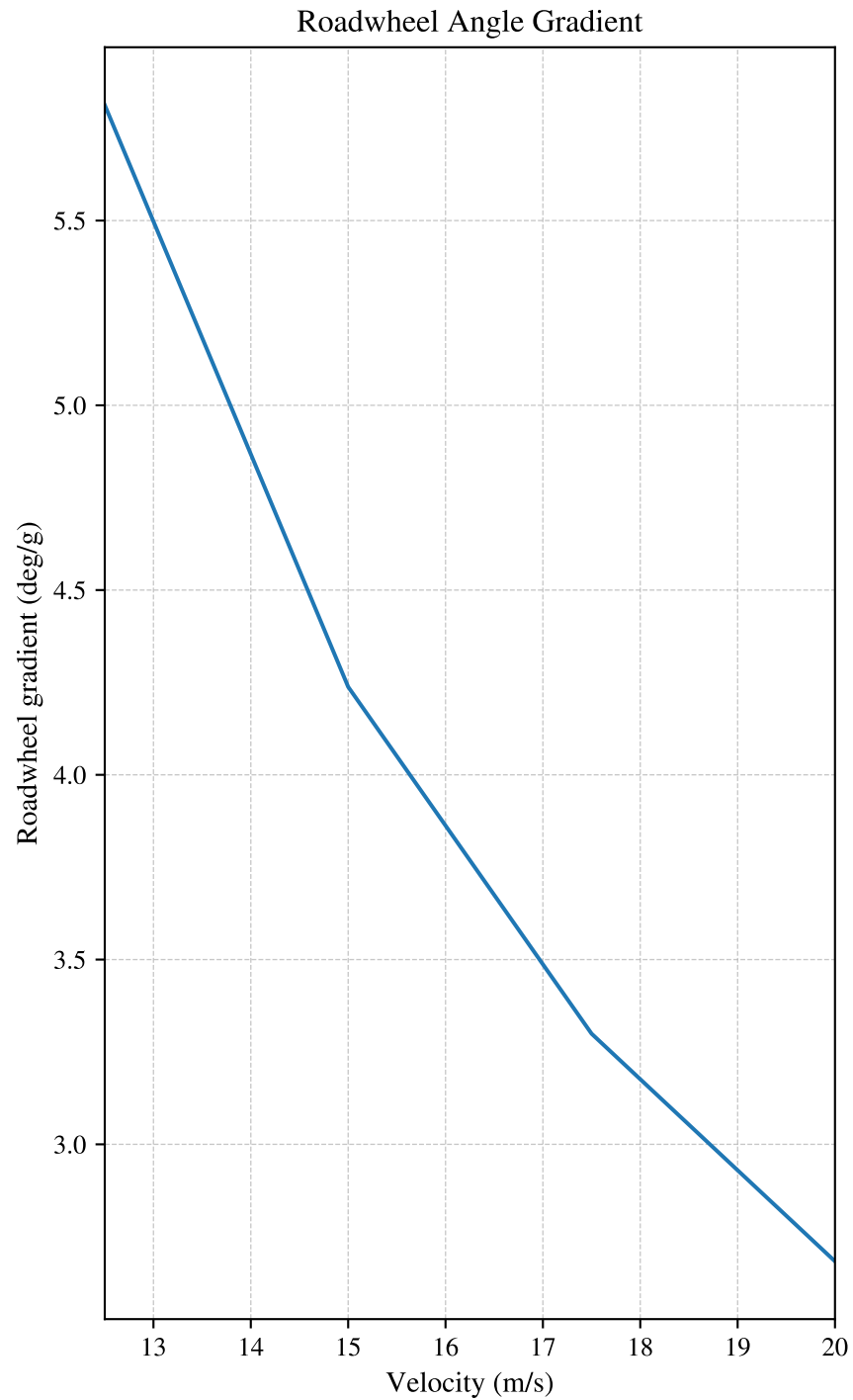
Steering Effort Isolines vs Measured a_y



Excess Steer Isolines vs Measured a_y

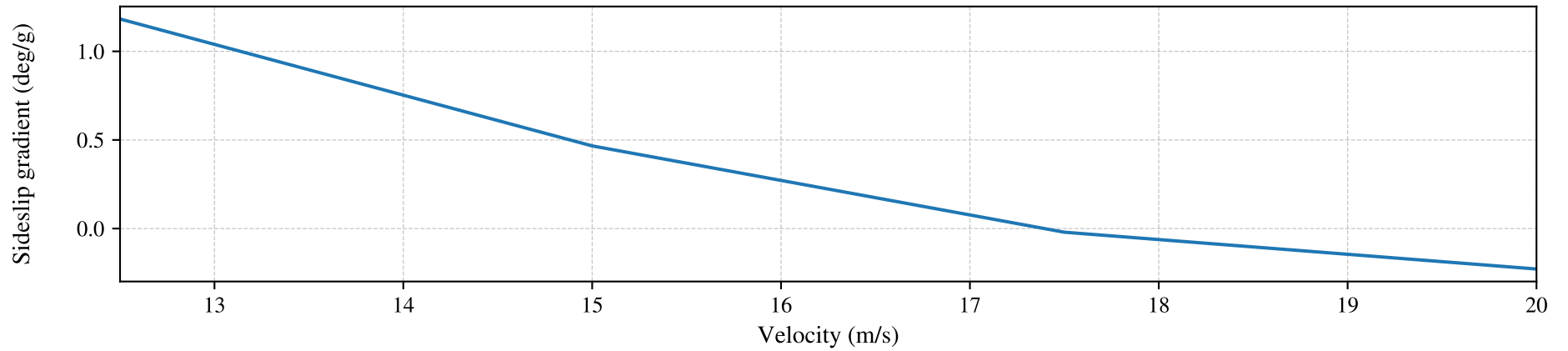


Angle Gradients vs Velocity

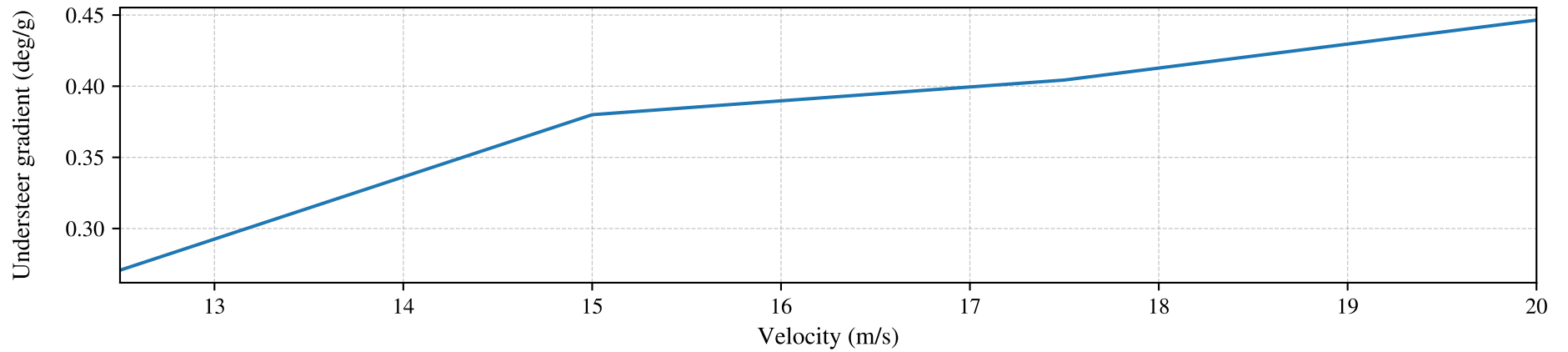


State Gradients vs Velocity

Sideslip Gradient



Understeer Gradient



Roll Gradient

